INCH-POUND
MIL-DTL-24643/45E
22 August 2002
SUPERSEDING
MIL-C-24643/45D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor, tin coated, ASTM B 286, AWG Size 26,7 strands.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation. Nominal wall thickness 0.015 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Telephone identification code shall be applied by method 3.
Fourth -	Two conductors cabled together with a maximum lay of 1-1/2 inches to form a pair.
Fifth -	The specified number of pairs (see table I) cabled together with a lay not greater than 16 times the overall diameter of the layer. Cabling sequence shall be consecutive, starting with No. 1, from the center outward.
Sixth -	Binder tape over each pair, applied at manufacturer's option, applied helically with overlap.
Seventh -	Braided shield of tin coated copper.
Eighth -	Cross-linked polyolefin jacket (see table I for thickness).

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TABLE I. Details.

Military part no. M24643/45	Size	No. of pairs	Cable jacket thickness (min avg) (inch)	Dia over shielded core (inch nom)	Overall diameter (inches) TYPE LS2U	
					min	max
-01UO	10	10	0.050	0.336	0.450	0.480
-02UO	15	15	0.050	0.393	0.530	0.560
-03UO	19	19	0.050	0.423	0.550	0.580
-04UO	30	30	0.050	0.545	0.670	0.700
-05UO	45	45	0.050	0.670	0.830	0.870
-06UO	60	60	0.065	0.750	0.920	0.960

EXAMINATION AND TESTS: Requirements

Basic electrical:
- Conductor resistance - ohms/1000 feet at 25°C, maximum 45.19
- Voltage withstand - volts, root mean square, minimum
  - Conductor to conductor 2000
  - Conductor to shield 1000
- Insulation resistance - Megohms -1000 feet minimum
  - Conductor to conductor 500
  - Conductor to shield 500
- Jacket flaws ..... No failure
- Conductor and shield continuity ..... No failure

Group A:
- Visual and dimensional ..... No failure
- Characteristic impedance - at 1MHz, ohms ..... 135 ± 15

Group B:
- Cross-linked proof test (percent, maximum)
  - Insulation 50
  - Jacket 50
- Cold bending, cable mandrel diameter 12 times cable diameter ..... No damage
- Bending endurance - at 25 ± 2°C, cycles, minimum 25
- Tear - pounds per inch thickness, minimum (ASTM D 470) 35

Physicals (unaged)
- Insulation
  - Tensile strength - lb/in^2, minimum 1800
  - Elongation - percent, minimum 250
- Jacket (cable)
  - Tensile strength - lb/in^2, minimum 1300
  - Elongation - percent, minimum 160

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Group C:
Physicals (aged)
- Insulation
  - Air oven
    - Tensile strength - percent of unaged, minimum 80
    - Elongation - percent of unaged minimum 80
- Jacket (cable)
  - Air oven
    - Tensile strength - percent of unaged, minimum 60
    - Elongation - percent of unaged minimum 60
  - Hot oil immersion
    - Tensile strength - percent of unaged, minimum 50
    - Elongation - percent of unaged, minimum 50
- Shrinkage ..... No failure
- Heat distortion - percent of unaged, maximum 30
- Permanence of printing (jacket) - cycles, minimum 125
- Shield - conformance to material, construction and coverage ..... No failure
- Surface transfer impedance (type LS2U only)
  - Milliohms per meter, maximum 700
  - EMP response time - dB, minimum 60

Group D:
- Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable)(125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50

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Smoke index, maximum
- Jacket 25
- Fillers 45
- Insulation 45

Toxicity index, maximum
- Jacket 5
- Fillers 5
- Insulation 1.5

Durometer (jacket) - (type A) hardness, minimum.....80
Weathering (jacket)..... No failure

UNIT ORDERING LENGTHS: All sizes 1000 feet (nominal)

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-043)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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